

21-3d Income and Substitution Effects

The impact of a change in the price of a good on the quantities purchased can be decomposed into two effects: an

income effect (the change in consumption that results when a price change moves the consumer to a higher or lower indifference curve)

and a

substitution effect (the change in consumption that results when a price change moves the consumer along a given indifference curve to a point with a new marginal rate of substitution)

. To see what these two effects are, consider how our consumer might respond when she learns that the price of Pepsi has fallen. She might reason in the following ways:

- “Great news! Now that Pepsi is cheaper, my income has greater purchasing power. I am, in effect, richer than I was. Because I am richer, I can buy both more pizza and more Pepsi.” (This is the income effect.)
- “Now that the price of Pepsi has fallen, I get more liters of Pepsi for every pizza that I give up. Because pizza is now relatively more expensive, I should buy less pizza and more Pepsi.” (This is the substitution effect.)

Which statement do you find more compelling?

In fact, both of these statements make sense. The decrease in the price of Pepsi makes the consumer better off. If pizza and Pepsi are both normal goods, the consumer will want to spread this improvement in her purchasing power over both goods. This income effect tends to make the consumer buy more pizza and more Pepsi. Yet at the same time, consumption of Pepsi has become less expensive relative to consumption of pizza. This substitution effect tends to make the consumer choose less pizza and more Pepsi.

Now consider the result of these two effects working at the same time. The consumer certainly buys more Pepsi because the income and substitution effects both act to increase consumption of Pepsi. But for pizza, the income and substitution effects work in opposite directions. As a result, whether the consumer buys more or less pizza is not clear. The outcome could go either way, depending on the relative magnitudes of the income and substitution effects. [Table 1](#) summarizes these conclusions.

Table 1

Income and Substitution Effects When the Price of Pepsi Falls

Good	Income Effect	Substitution Effect	Total Effect
Pepsi	Consumer is richer, so she buys more Pepsi.	Pepsi is relatively cheaper, so consumer buys more Pepsi.	Income and substitution effects act in the same direction, so consumer buys more Pepsi.
Pizza	Consumer is richer, so she buys more pizza.	Pizza is relatively more expensive, so consumer buys less pizza.	Income and substitution effects act in opposite directions, so the total effect on pizza consumption is ambiguous.

We can interpret the income and substitution effects using indifference curves. *The income effect is the change in consumption that results from the movement to a new indifference curve. The substitution effect is the change in consumption that results from moving to a new point on the same indifference curve with a different marginal rate of substitution.*

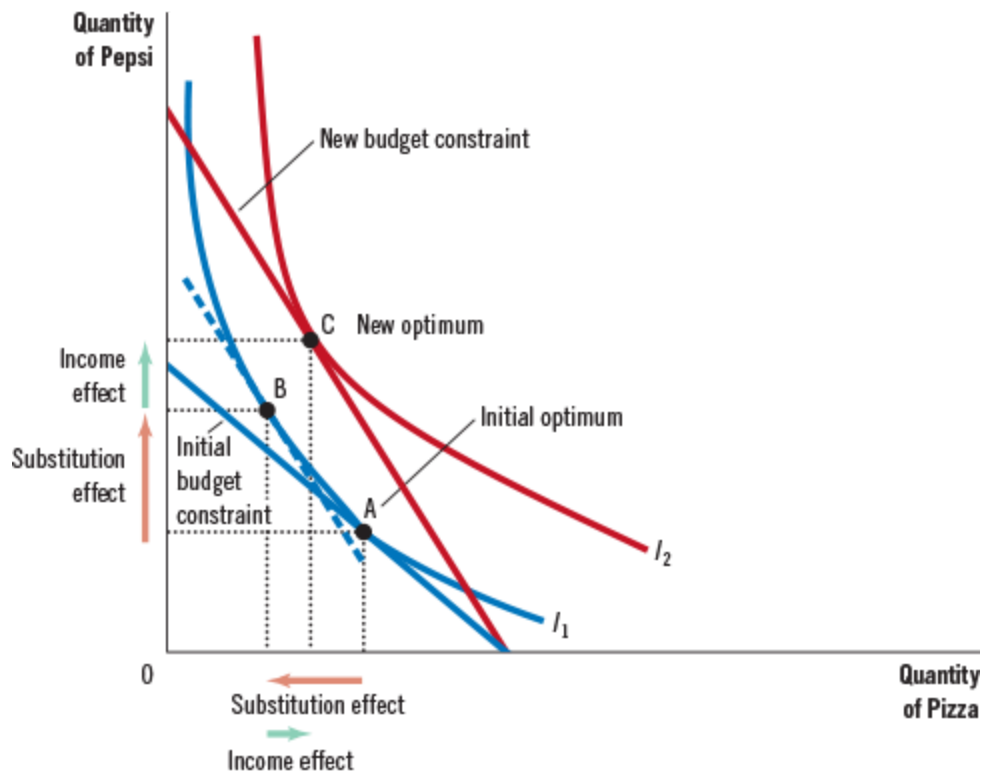
Figure 11 shows graphically how to decompose the change in the consumer's decision into the income effect and the substitution effect. When the price of Pepsi falls, the consumer moves from the initial optimum, point A, to the new optimum, point C. We can view this change as occurring in two steps. First, the consumer moves *along* the initial indifference curve, I_1 , from point A to point B. The consumer is equally happy at these two points, but at point B, the marginal rate of substitution reflects the new relative price. (The dashed line through point B is parallel to the new budget constraint and thus reflects the new relative price.) Next, the consumer *shifts* to the higher indifference curve, I_2 , by moving from point B to point C. Even though point B and point C are on different indifference curves, they have the same marginal rate of substitution. That is, the slope of the indifference curve I_1 at point B equals the slope of the indifference curve I_2 at point C.

Figure 11

Income and Substitution Effects

The effect of a change in price can be broken down into an income effect and a substitution effect. The substitution effect—the movement along an indifference curve to a point with a different marginal rate of substitution—is shown here as the change from point A to point B along indifference curve I_1 . The income effect—the

shift to a higher indifference curve—is shown here as the change from point B on indifference curve I_1 to point C on indifference curve I_2 .



The consumer never actually chooses point B, but this hypothetical point is useful to clarify the two effects that determine the consumer's decision. Notice that the change from point A to point B represents a pure change in the marginal rate of substitution without any change in the consumer's welfare. Similarly, the change from point B to point C represents a pure change in welfare without any change in the marginal rate of substitution. Thus, the movement from A to B shows the substitution effect, and the movement from B to C shows the income effect.